

Sadia Afroz Oishi

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Oishi

OVERVIEW

Graduate B.Sc. (Engineering) in Information and Communication Engineering from Pabna University of Science and Technology with a CGPA of 3.67 / 4.00 (all 8 semesters completed). Research experience focuses on Artificial Intelligence, deepfake forensics and digital integrity, computer vision, deep learning, and AI applications in smart agriculture. First author and co-author of peer-reviewed IEEE conference publications, submitted high-impact journal manuscripts, a published open-access Mendeley Data dataset, and active scholarly peer reviewer for international IEEE conferences.

EDUCATION

- 2022–2026 B.Sc. (Engineering) in Information and Communication Engineering**
Pabna University of Science and Technology (PUST), Bangladesh · CGPA: 3.67 / 4.00 (All 8 Semesters Completed · Final Result Published)
Medium of Instruction: English (100% official instructional medium across all coursework and thesis defenses).
Undergraduate Thesis: "Auto-IFCB-STNet: A Unified Spatio-Temporal Forensic Concept Bottleneck for Explainable Image and Video Deepfake Detection." Submitted to *Scientific Reports* (Nature Portfolio).
- 2018–2020 Higher Secondary Certificate (HSC) — Science**
Savar Model College, Savar, Dhaka · GPA: 5.00 / 5.00 (Golden Merit) · Dhaka Board
- 2016–2018 Secondary School Certificate (SSC) — Science**
Radio Colony Model School and College, Savar, Dhaka · GPA: 5.00 / 5.00 (Golden Merit) · Dhaka Board

ACADEMIC SERVICE & PEER REVIEW

- **Peer Reviewer, 5th IEEE RAAICON International Conference (2026)** — Designated peer reviewer evaluating scholarly submissions in computational intelligence, neural networks, and pattern recognition. Received official Certificate of Appreciation.
- **Peer Reviewer, 4th IEEE RAAICON International Conference (2025)** — Evaluated submitted research manuscripts in machine learning, computer vision, and AI architectures at MIST, Dhaka. Received official Certificate of Appreciation.

PUBLICATIONS & SCHOLARLY RECORDS

Peer-Reviewed Conference Papers:

1. **S. A. Oishi**, A. Islam, M. A. Miya, M. H. Mohona, M. S. Hossain et al., "A Comparative Study of CNN, Transformer, and Recurrent Models for Deepfake Detection on Degraded Images," *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII)*, IEEE, 2026. [IEEE Xplore: 11662017](#) **[First Author]**.
2. A. Islam, **S. A. Oishi**, M. M. Hasan, M. Mamun, M. A. Hossain, M. H. Mohona, M. A. Miya, S. K. Ray, "GR-ACE Net: A Hybrid Graph-Attentional Framework with Global Relational Reasoning for Deepfake Forensics," *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII)*, IEEE, 2026. [IEEE Xplore: 11661873](#) **[Second Author]**.

3. A. Islam, S. K. Ray, **S. A. Oishi** et al., "A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, 2026. DOI: [10.1109/QPAIN69676.2026.11545784](https://doi.org/10.1109/QPAIN69676.2026.11545784).
4. M. A. Miya, M. H. Mohona, M. A. Hossain, M. B. A. Z. Shammo, A. Islam, **S. A. Oishi** et al., "Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices," *2025 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, 2025. DOI: [10.1109/ICCIT68739.2025.11491546](https://doi.org/10.1109/ICCIT68739.2025.11491546).
5. **S. A. Oishi** et al., "SpatSeqFake: A Hybrid Spatial-Sequential Framework for Cross-Dataset Deepfake Image Detection," Presented at *2026 5th International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (RAAICON 2026)*, IEEE, 2026 **[First Author]**.
6. S. K. Ray, A. Islam, M. C. Chanda, **S. A. Oishi**, M. S. Hosain, "Deep Learning Based Lung Image Segmentation Using XR-U-Net," *2024 27th International Conference on Computer and Information Technology (ICCIT)*, IEEE, 2024. DOI: [10.1109/ICCIT64611.2024.11022346](https://doi.org/10.1109/ICCIT64611.2024.11022346).

Submitted Journal Manuscripts:

1. **S. A. Oishi** et al., "Auto-IFCB-STNet: A Unified Spatio-Temporal Forensic Concept Bottleneck for Explainable Image and Video Deepfake Detection," Submitted to *Scientific Reports* (Nature Portfolio) **[First Author]**.
2. A. Islam, **S. A. Oishi**, M. M. Hasan et al., "Development and Evaluation of a YOLO-Based Mango Fruit Health Detection Framework Using the MangoFruitBD Dataset," Submitted to *Computers and Electronics in Agriculture* (Elsevier, Q1; CiteScore: 17.3) **[2nd Co-Author]**.
3. S. K. Ray, A. Islam, A. Kumar, **S. A. Oishi**, M. A. Miya, M. A. Hossain*, "Transfer Learning Optimization for Mango Fruit Disease Classification: A Benchmark Study on a Novel Bangladeshi Dataset," Submitted to *Annals of Emerging Technologies in Computing* [Q2; CiteScore: 4.0].

Published Research Dataset:

1. A. Islam, **S. A. Oishi**, M. M. Hasan et al., "MangoFruitBD: A Bounding-Box Annotated Image Dataset for Detecting Healthy and Diseased Mango Fruits in Bangladeshi Orchards," *Elsevier Mendeley Data*, Version 1 & 2, 2026. DOI: [10.17632/bhrz29mkmr.1](https://doi.org/10.17632/bhrz29mkmr.1) **[2nd Co-Author & Annotation Engineer]**.

SELECTED RESEARCH EXPERIENCE

Explainable Deepfake Forensics & STNet

2025–2026

- Formulated spatio-temporal concept bottleneck architectures (Auto-IFCB-STNet and SpatSeqFake) for explainable image and video deepfake detection across multi-generator manipulation benchmarks.
- Investigated artifact retention dynamics under severe degradation (compression, noise, blur) using CNN, Vision Transformer, and Recurrent frameworks.

MangoFruitBD & YOLO-Based Mango Health Detection

2025–2026

- Co-developed field-collected dataset of 1,310 RGB orchard images across Healthy, Anthracnose, Alternaria, and Scab classes.
- Engineered bounding-box annotations using CVAT and Roboflow; evaluated state-of-the-art YOLO detectors for real-time orchard disease diagnosis.

TECHNICAL & RESEARCH SKILLS

Research & AI:	Deep Learning, Computer Vision, Deepfake Forensics, Explainable AI (XAI), Concept Bottlenecks, Image & Video Analysis, YOLO Object Detection, Agricultural AI, Model Training & Evaluation.
Data Curation:	CVAT (Computer Vision Annotation Tool), Roboflow, LabelImg, Field Data Acquisition, Dataset Auditing, Augmentation.
Programming:	Python, C, C++, SQL, JavaScript, MATLAB, HTML5 / CSS3.
Tools & Platforms:	Visual Studio Code, Jupyter Notebook, Google Colab Pro, Git & GitHub, MS SQL Server, Proteus, Cisco Packet Tracer, LaTeX / Overleaf.
Coursework:	Artificial Intelligence & Robotics, Neural Networks, Natural Language Processing, Digital Image & Speech Processing, DBMS, Software Analysis & System Testing, Wireless

ACHIEVEMENTS & HONORS

- **Debate Champion & Debater of the Tournament (2025):** Grand Champion with Team Voice of Victory at the ICE Fiesta Parliamentary Debate Tournament, and awarded individual highest-speaker trophy.
- **Peer Review Recognition (2025 & 2026):** Awarded official certificates for scholarly peer-review service at 4th and 5th IEEE RAAICON international conferences.
- **Academic Distinction:** Achieved perfect GPA 5.00 / 5.00 in both Higher Secondary Certificate (HSC 2020) and Secondary School Certificate (SSC 2018).

TALKS & PRESENTATIONS

- **2026:** Presented "*SpatSeqFake: A Hybrid Spatial-Sequential Framework for Cross-Dataset Deepfake Image Detection*" as First Author at IEEE RAAICON 2026.
- **Jun 2026:** Presented "*A Comparative Study of CNN, Transformer, and Recurrent Models for Deepfake Detection on Degraded Images*" as First Author at IEEE PECCII 2026.
- **Apr 2026:** Presented "*A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition*" at IEEE QPAIN 2026.
- **Dec 2025:** Presented "*Remote Health Monitoring via PPG Signal Processing*" at IEEE ICCIT 2025.
- **2025:** Delivered oral research presentation at 1st RUEC International Conference, Rajshahi University.

LEADERSHIP & EXTRA-CURRICULAR

- **Joint Organizing Secretary, PUST Career & Entrepreneurship Club (PUSTCEC) (2025–2026):** Coordinated club career seminars, freshers' reception, and skill workshops.
- **Founding Member, PUSTCEC (2024–Present):** Formally signed the Founding Charter of PUSTCEC.
- **Podcast Host & Producer (2025):** Hosted Episode 02 of the official PUSTCEC "Mic & Minds" podcast.
- **Master of Ceremonies (2025):** Official stage anchor for 1st ICE Alumni Reunion (Formal Ceremony & Cultural Concert).

REFERENCES

Dr. Md. Sarwar Hossain

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